

Biomechanical principles of tooth preparation

lecture four

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Principles of tooth preparation:

1-Preservation of the tooth structure:

The preparation must be conservative, the minimal amount of tooth structure must be removed, excessive reduction (cutting) will be destructive and has many harmful effects:-

- 1- Lead to thermal hypersensitivity, pulpal inflammation and necrosis (death of the pulp) may result.
- 2- The tooth might be over tapered or shortened and this may have adverse effect on the retention and resistance of the crown on prepared tooth.



Excessive tooth reduction: The tooth is over tapered and shortened and this will affect the retention and resistance of the prepared tooth.

Many techniques may be useful in prepare a tooth in conservative way:

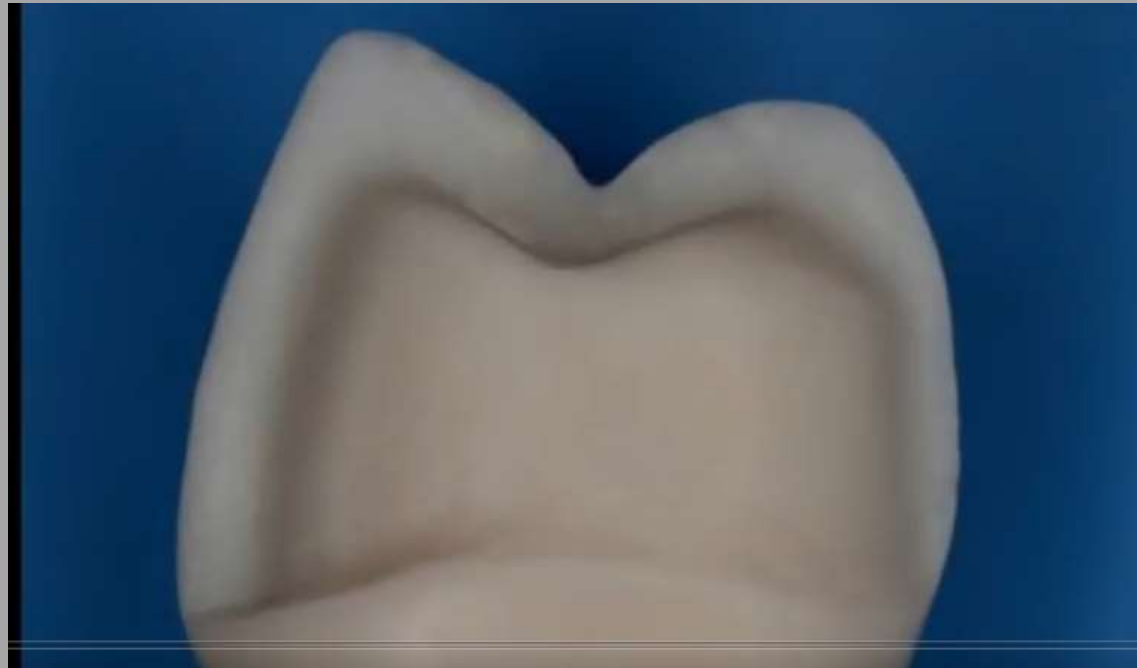
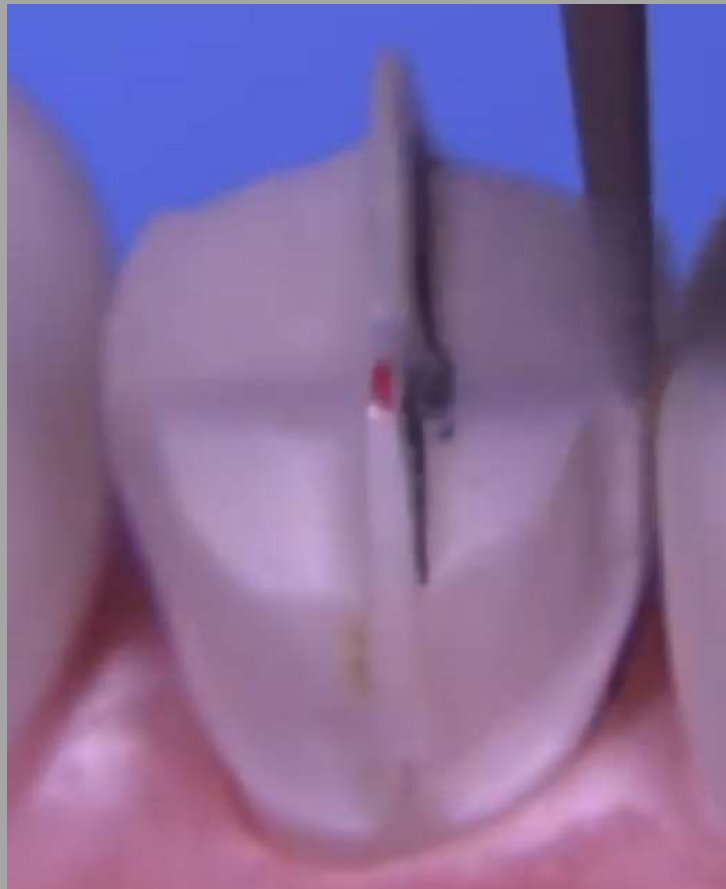
first, controlling the amount of tooth structure removed by:

Using depth orientation grooves will help in controlling the amount of reduction.



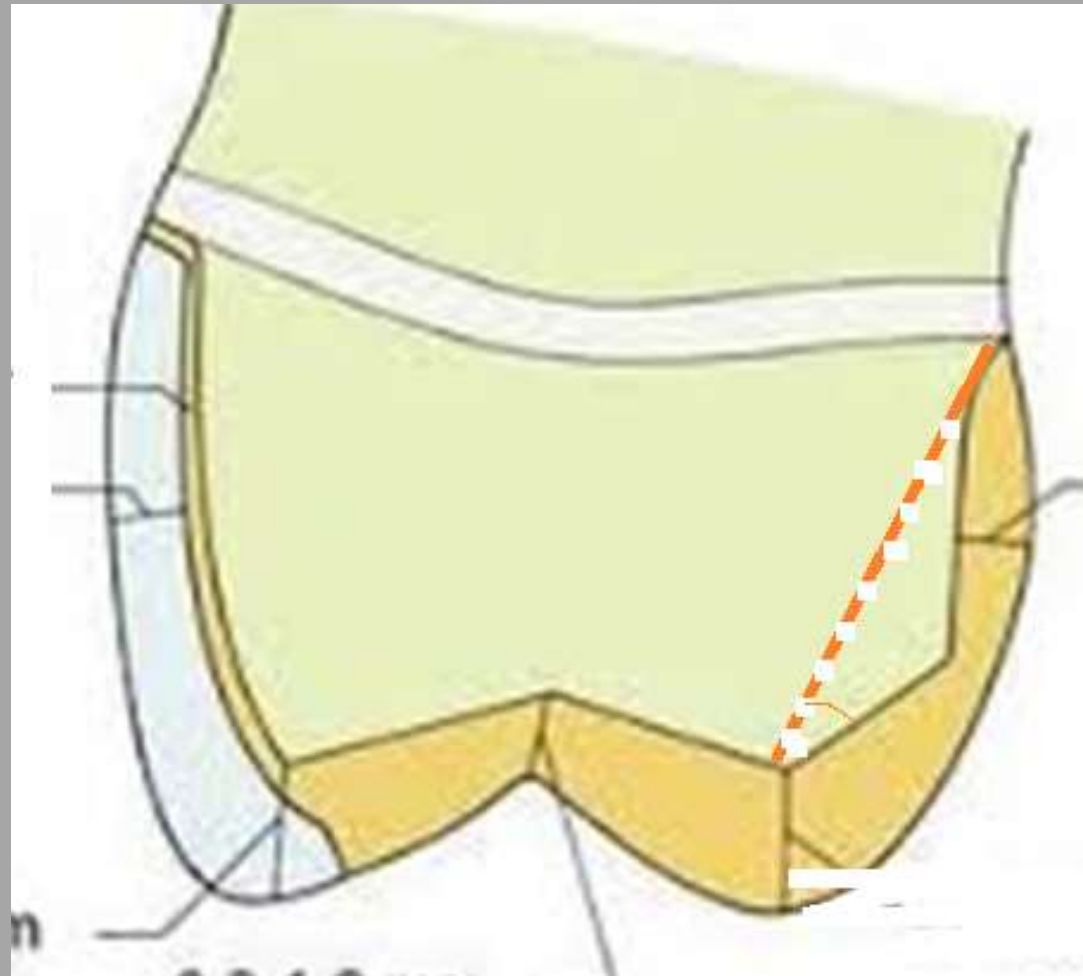
Second: Tenon preparation

It is done by preparing a surface removing the undercuts and following the normal anatomical irregularities leaving a tenon for measuring the amount of tooth removed.



T-PREP*
Technique

Third : two-step reduction will maintain the anatomical feature of prepared tooth surface for more tooth conservation and more retention.



Fourth: mold technique

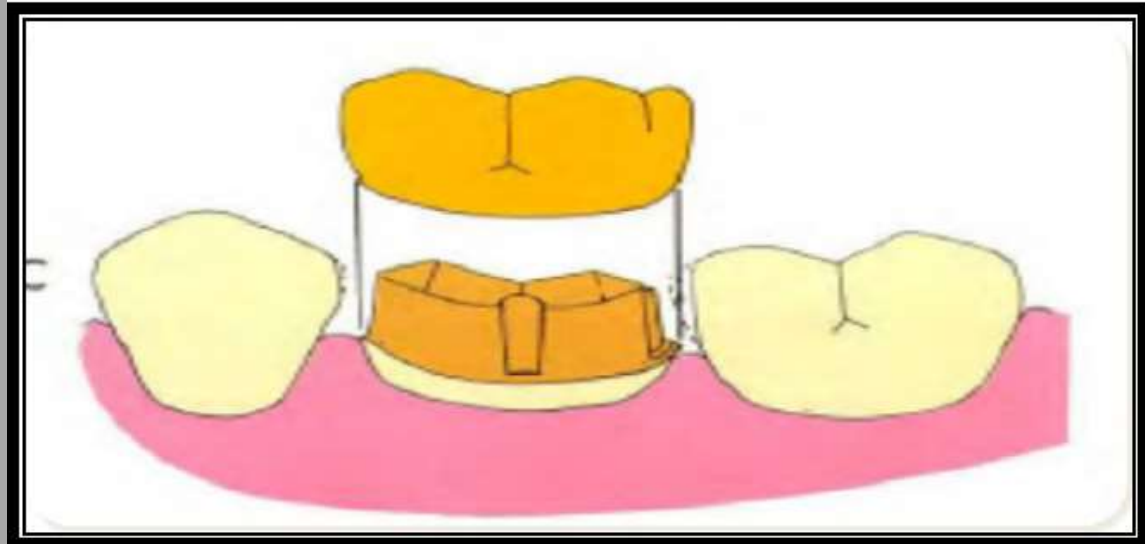
This done by making mold by heavy body of impression material before preparation and observe the amount of tooth removed by returning the mold on prepared tooth



2- Retention and resistance form:

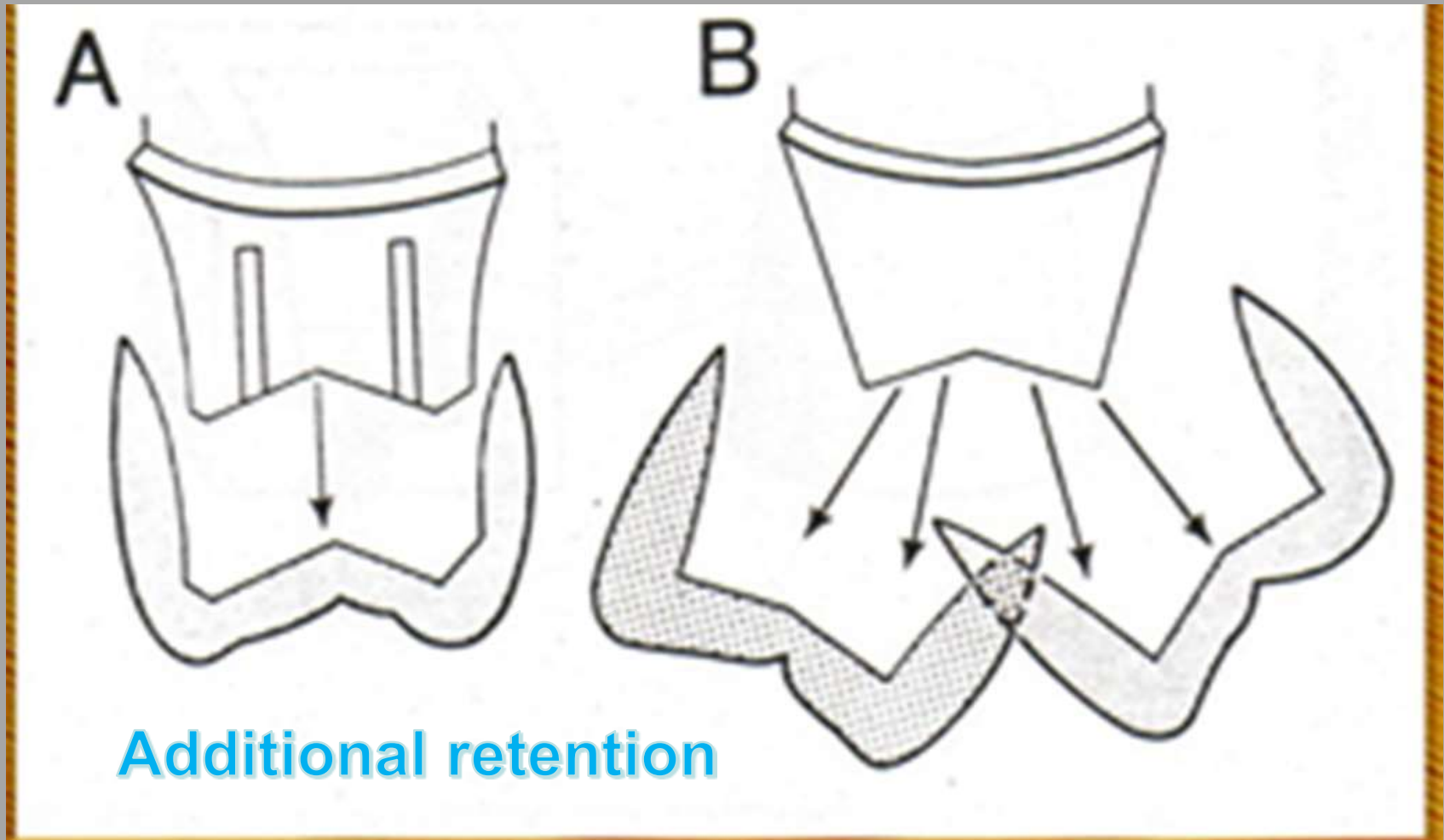
Path of insertion:

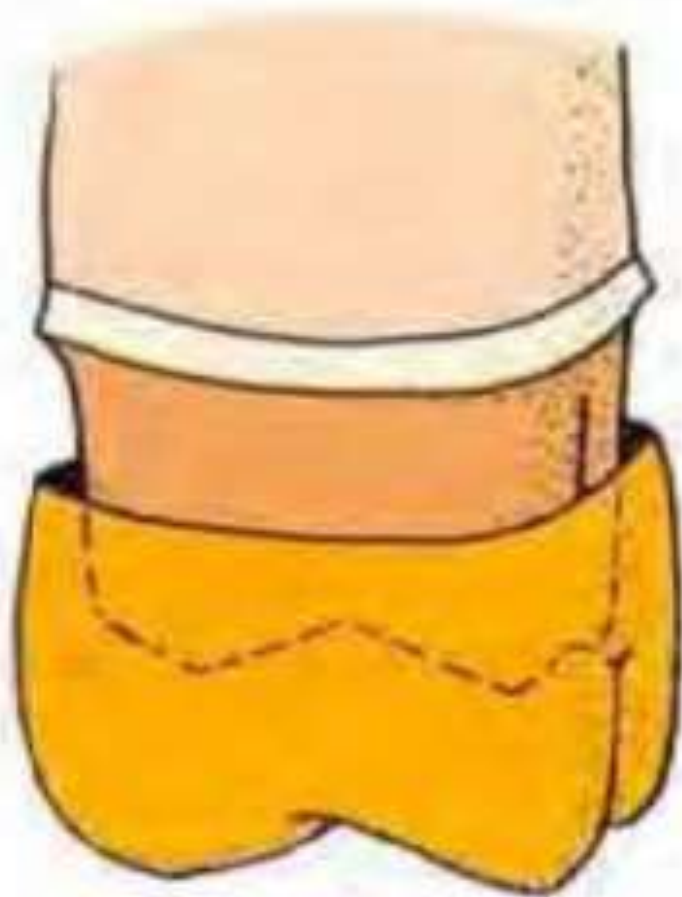
An imaginary line along which the restoration can be inserted and removed without causing lateral force on the abutment.



Path of insertion:

The crown restoration should have a single path of insertion to be retentive.

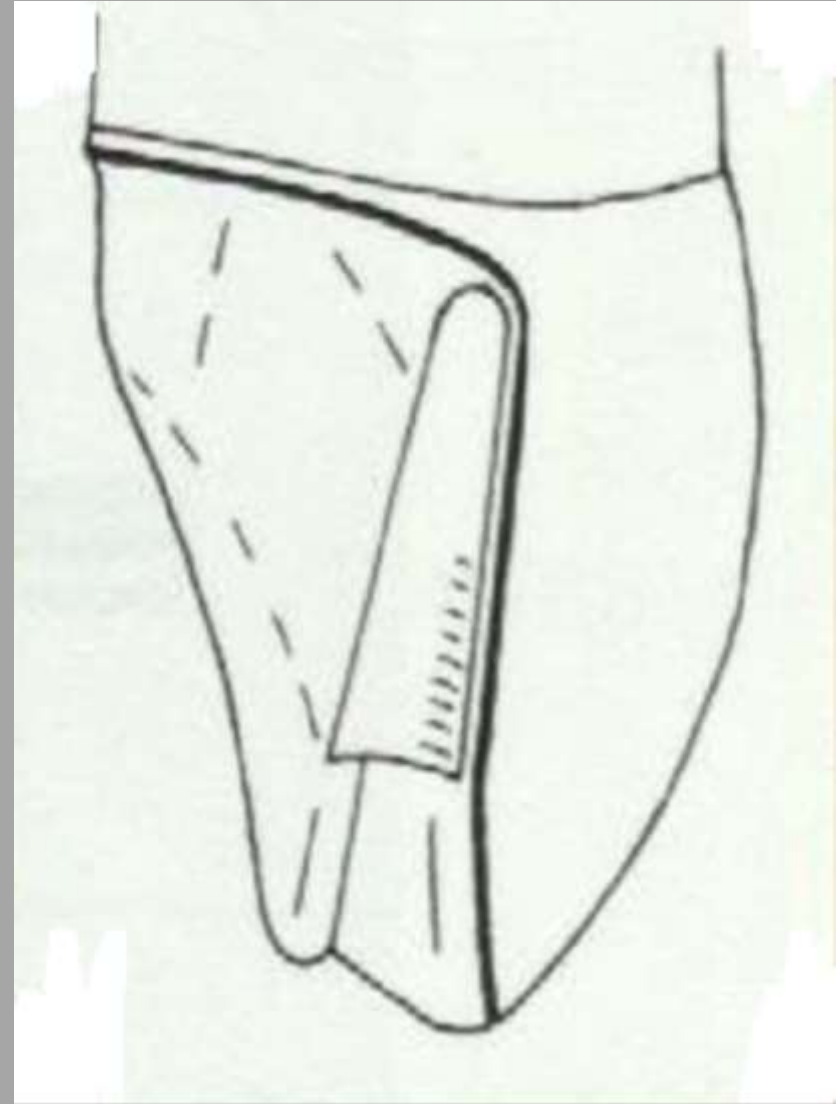




**Avoid round section preparation
and keep anatomical features**

Path of insertion:

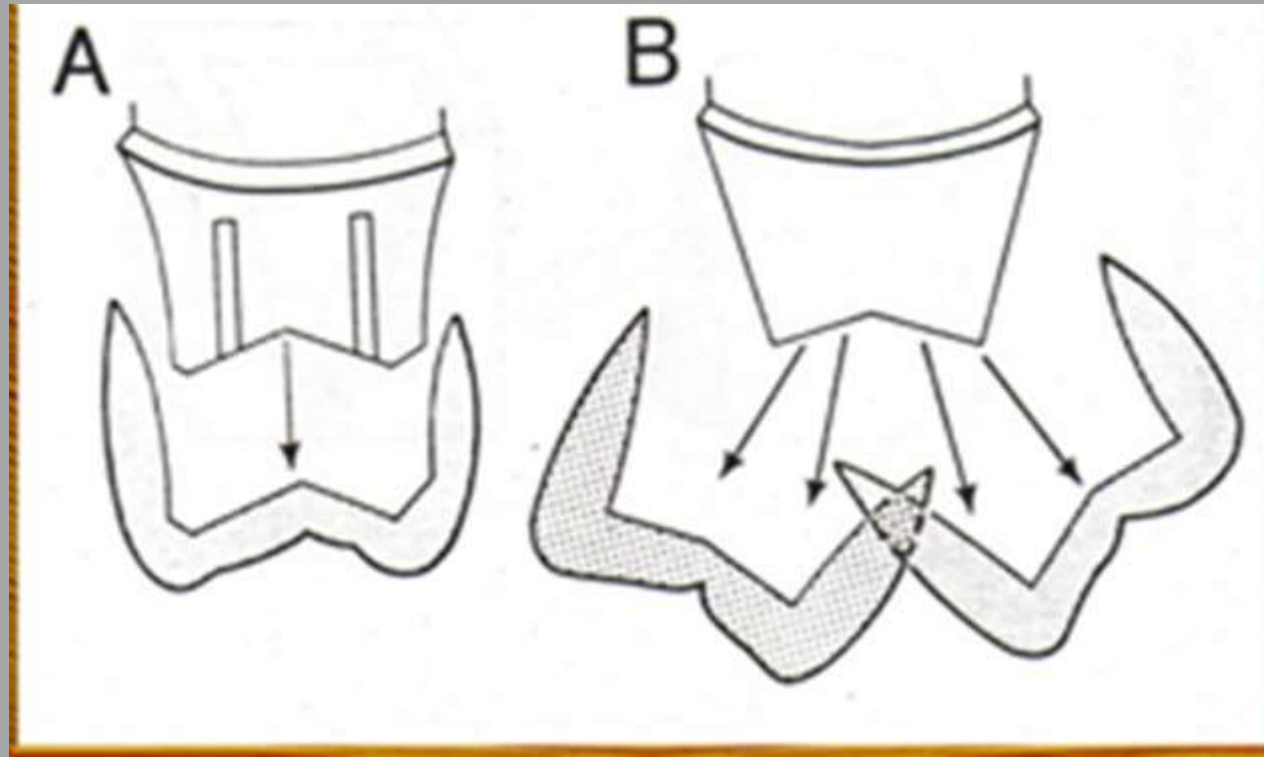
Most of the time, the path of insertion of the crown restoration is parallel to the long axis of the tooth, but this is not a rule as in three-quarter crown for the anterior teeth where the path of insertion should be parallel to the incisal two-thirds of the crown not to the long axis.



2- Retention and resistance form:

Retention: is the ability of the preparation to resist the crown restoration removal along its path of insertion.

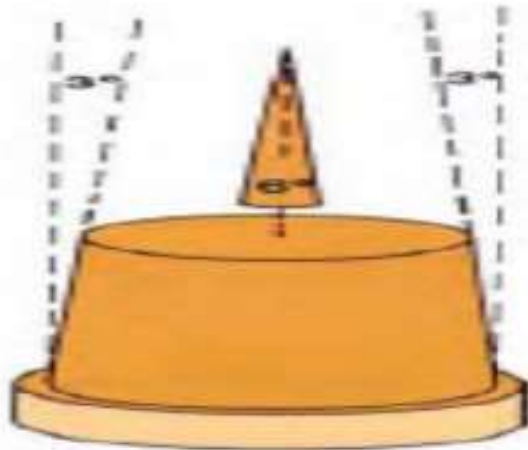
Resistance: is the ability of the preparation to resist the dislodgment of the restoration by forces directed obliquely or horizontally to the restoration.



Factors which have effect on retention and resistance:

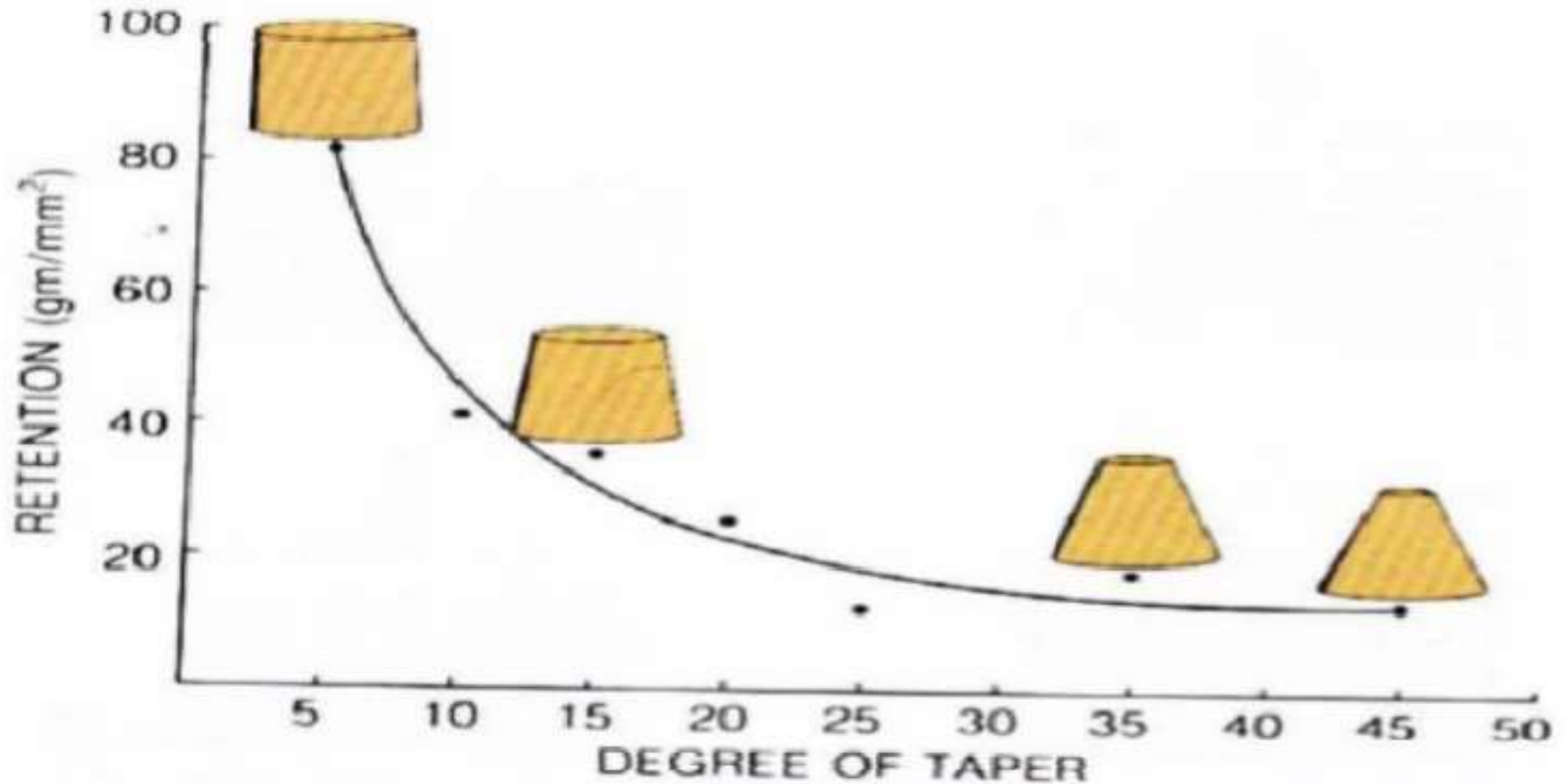
1. Taper of the preparation:

Convergence angle is the angle that is formed between each two opposing axial walls of a tooth prepared to receive a crown restoration. It determines the convergence (taper) of the prepared tooth.



Taper of the preparation:

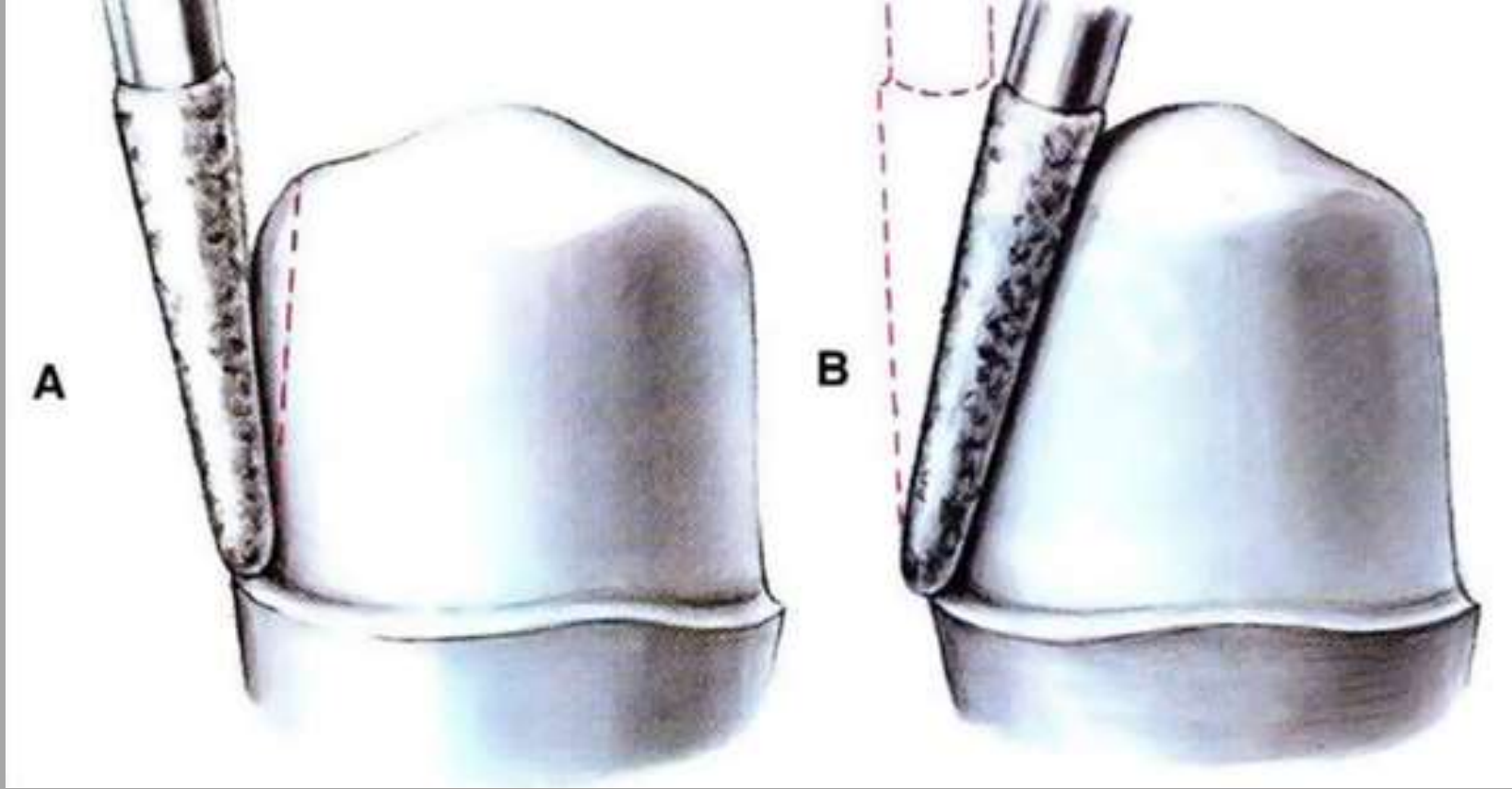
The magnitude of retention depends on the degree of this angle, the greater the taper the less the retention. The degree of the convergence angle is one of the factors that determine the amount of axial and non-axial forces which can be tolerated without leading to dislodgement of the crown restoration.



Inverse relationship

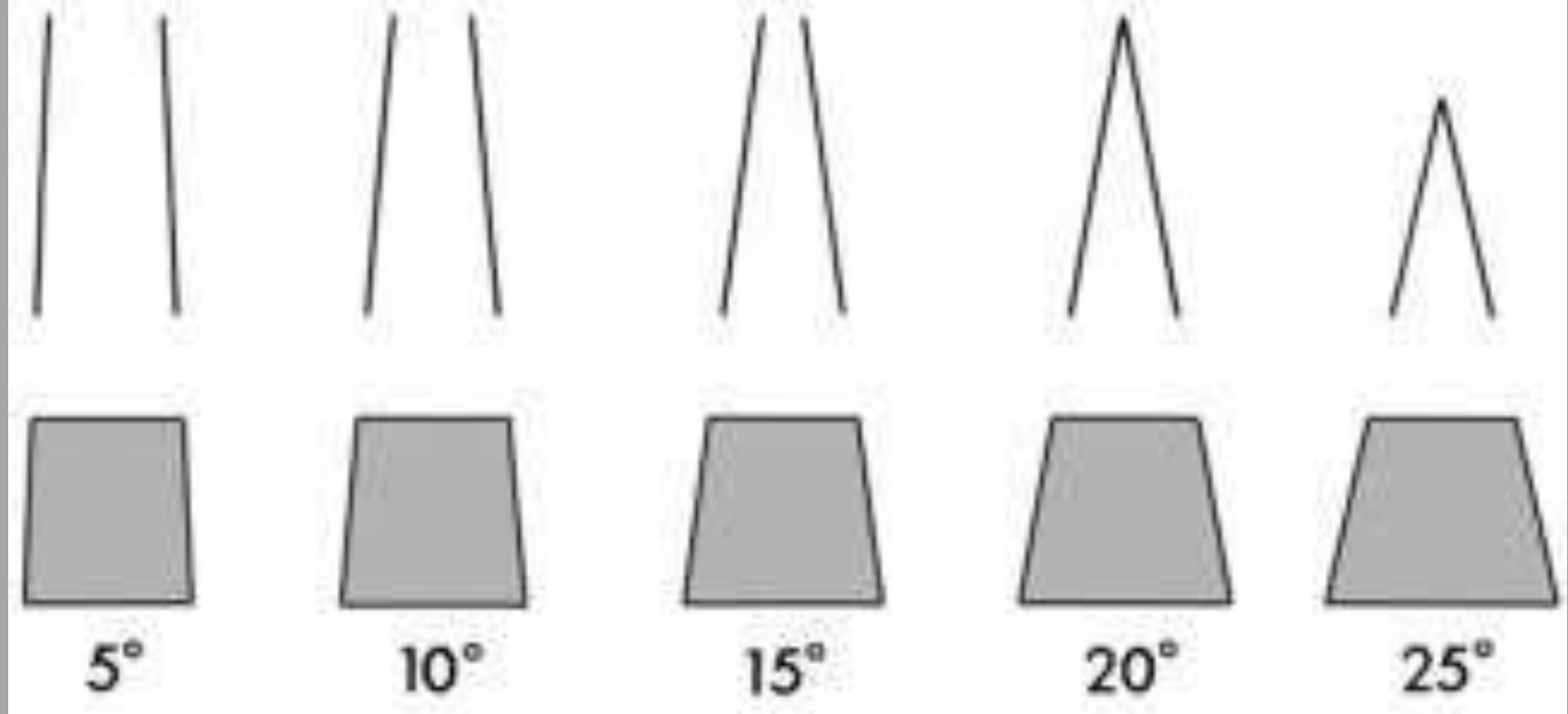
Taper of the prepared walls :

5-6 degrees convergence angle is mostly used to provide the needed retention. The more nearly parallel the opposing walls of preparation, the greater will be the retention, but parallel walls are difficult to be obtained inside the patient's mouth without creating undercuts and might lead to difficulty in seating of the crown restoration, thus 5-6 degrees convergence angle is mostly used to provide the needed retention.



At left (A) the diamond bur is tipped away from the path of insertion (long axis) resulting in an undercut.

At right (B) the bur is tipped into the tooth too far leading to excessively tapered preparation.

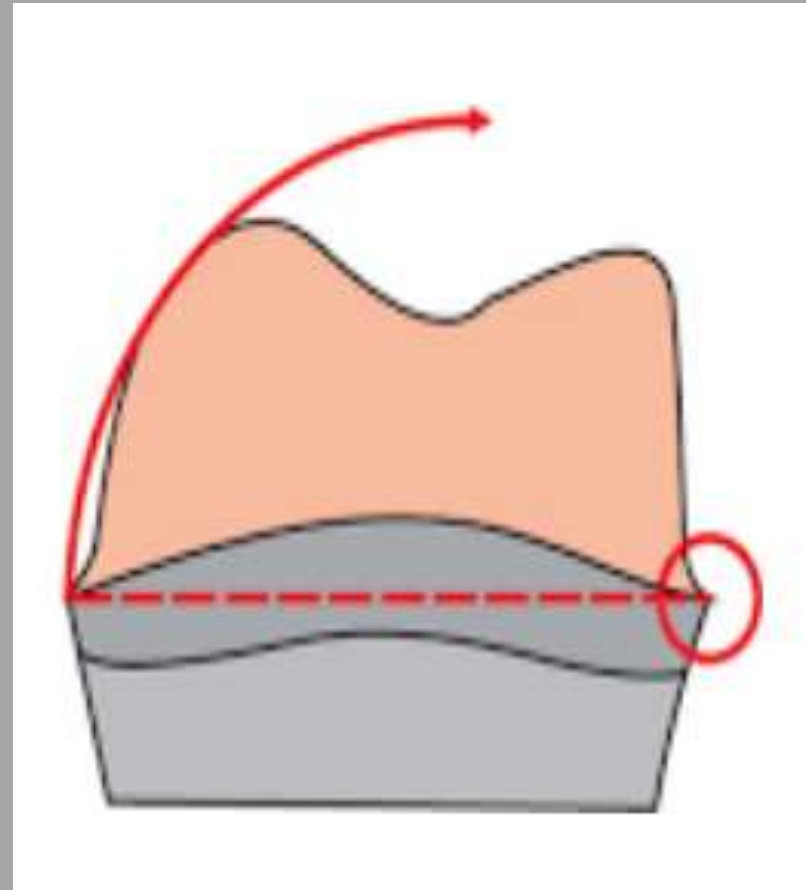


5-7° is the required amount of taper between opposite walls of prepared tooth .

NOTE: Retention will not be decreased gradually according to the increase in walls taper, it may be **lost** suddenly with slight increase in taper so it is considered as **critical factor**.

Taper and Resistance:

The more parallel the axial walls of the preparation, the more will be the resistance of crown restoration. The walls of a short wide preparation must be kept nearly parallel to achieve adequate resistance from.



2- Surface area of the preparation :
increasing the surface area lead to increased retention.

Mark desired position of mesial and distal retention/resistance grooves

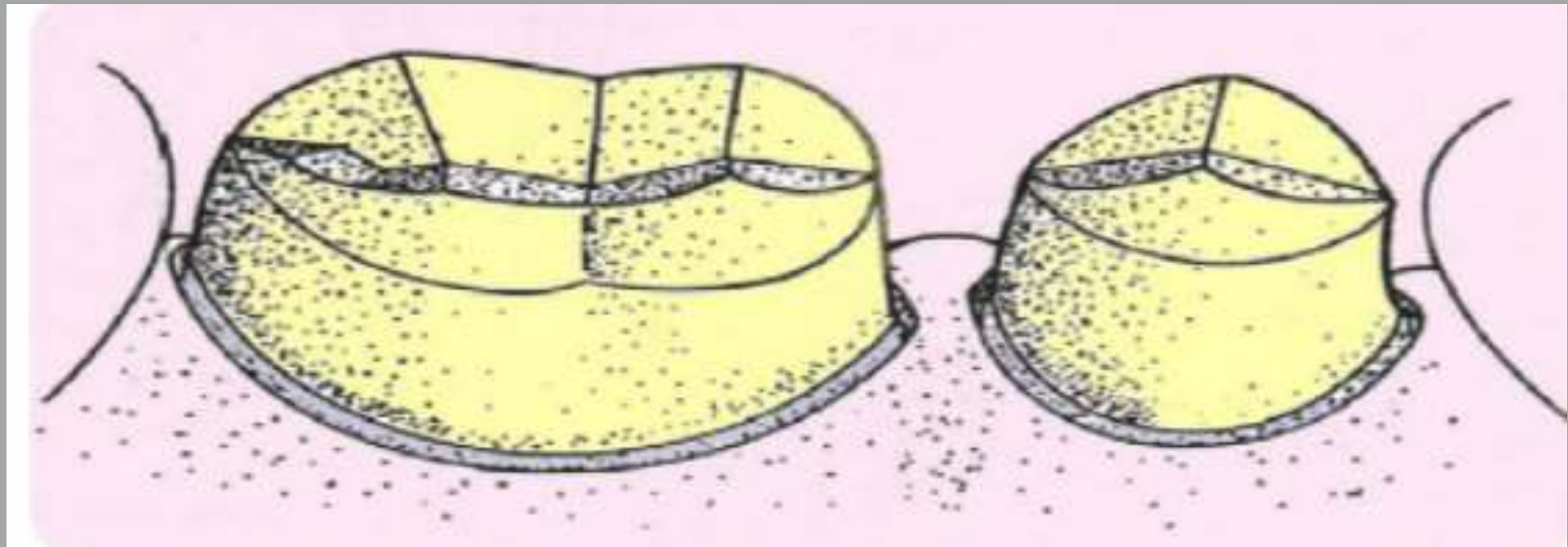


Suitable burs for grooves
+T.C set 1169
+330 short tapered diamond



The factors that influence the surface area are:

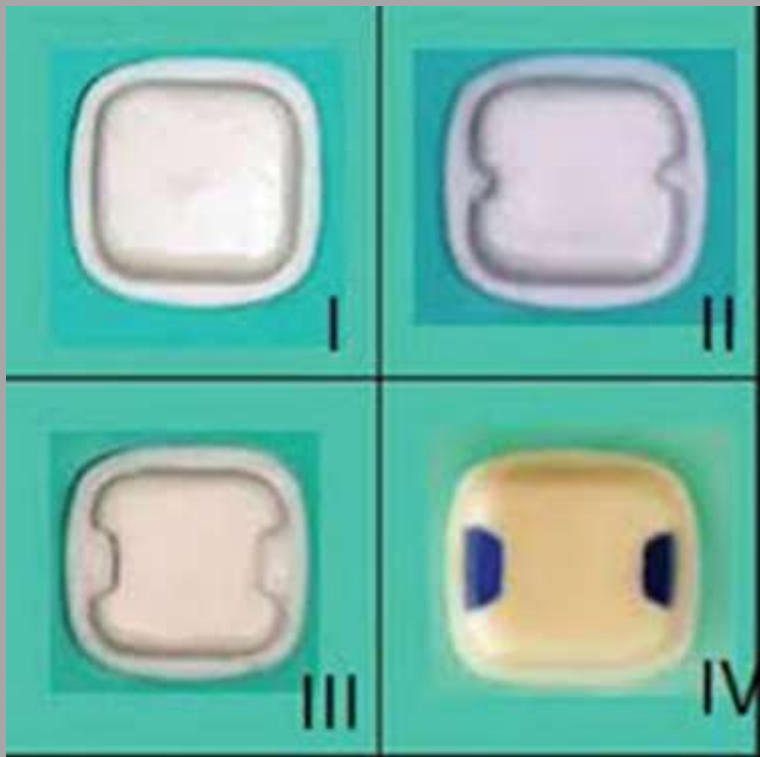
1- Size of the tooth: The larger the size of the tooth, the more will be the surface area of the preparation, and thus the more will be the retention. So a full metal crown on a molar tooth will be more retentive than that on a premolar tooth.



The factors that influence the surface area are:

2- Extent of tooth coverage by the restoration: The more the area that will be covered by the crown restoration, the more will be the retention. Thus full metal crown on a molar tooth is more retentive than a three-quarter crown on the same tooth.

3- Accessory features: such as boxes, grooves, and pin holes



3- Length of the prepared tooth:

Increasing the length of prepared tooth, retention will be increased.

Increasing length may be done by:

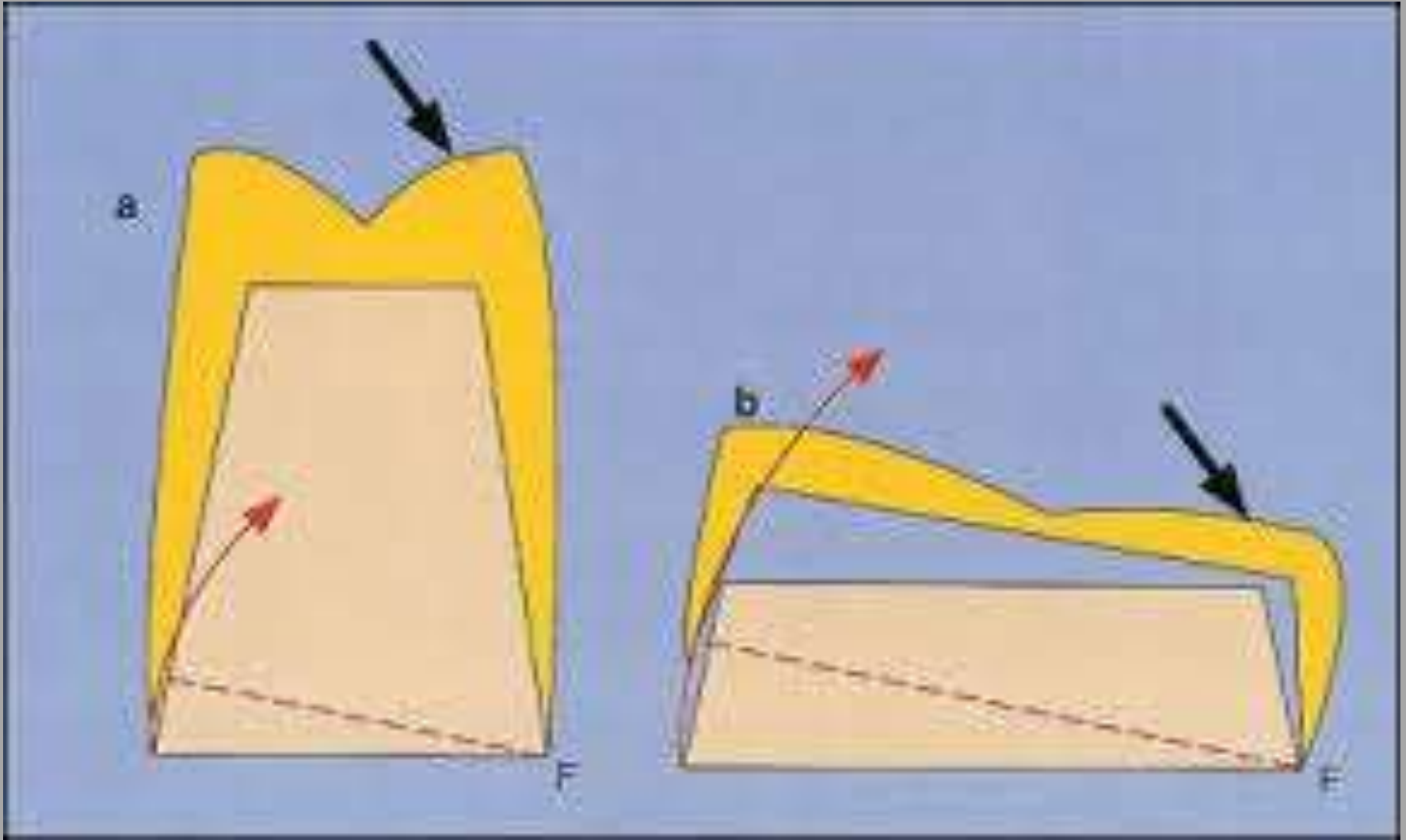
1- using subgingival finishing line.

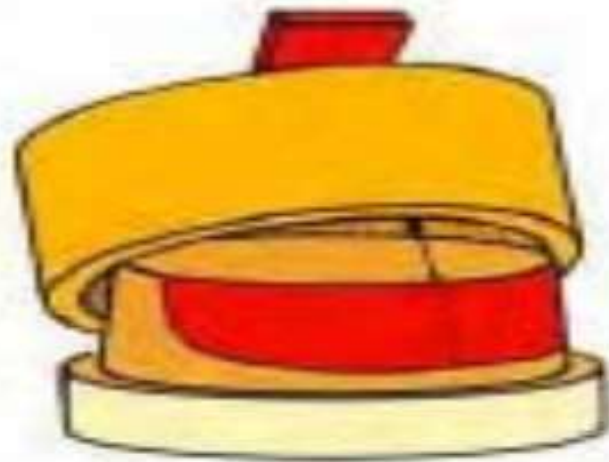
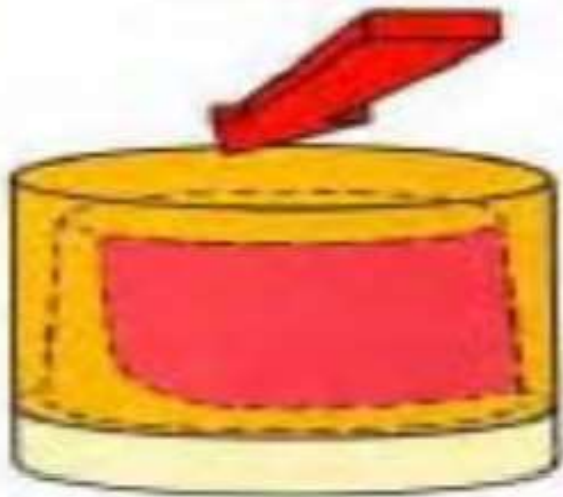
2- surgical removal of gingival tissue using laser surgery (crown lengthening technique).



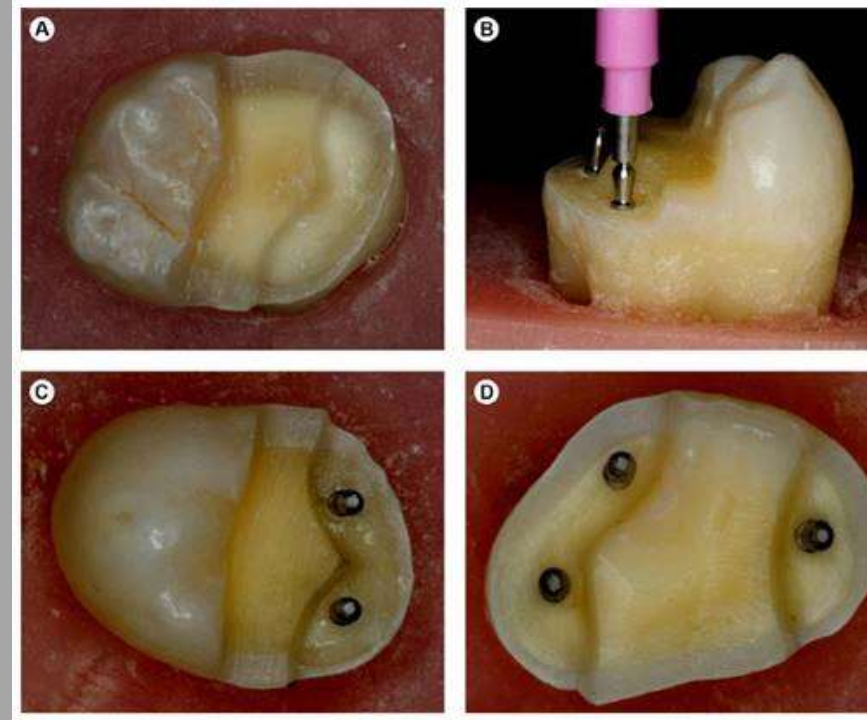
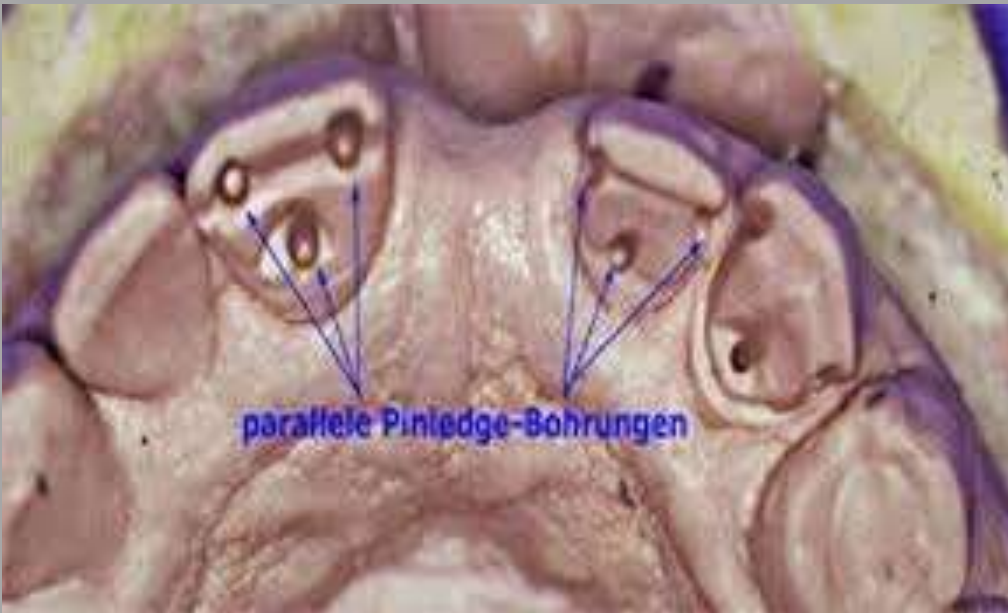
Length of the prepared tooth:

Increasing the length of the preparation there will be increase in the retention and resistance and vice versa.

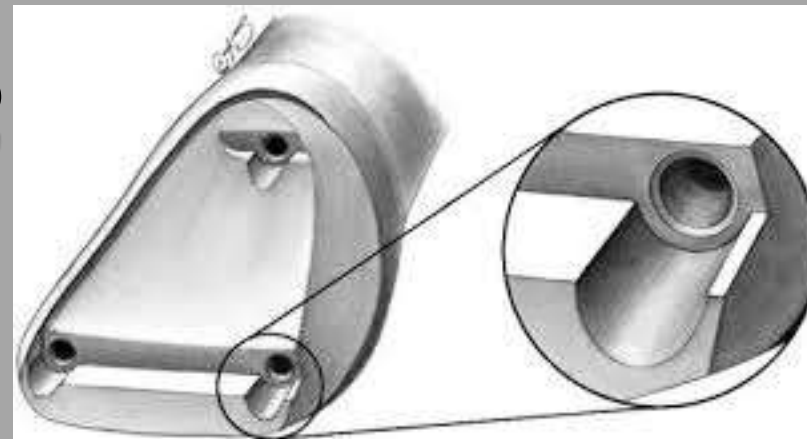




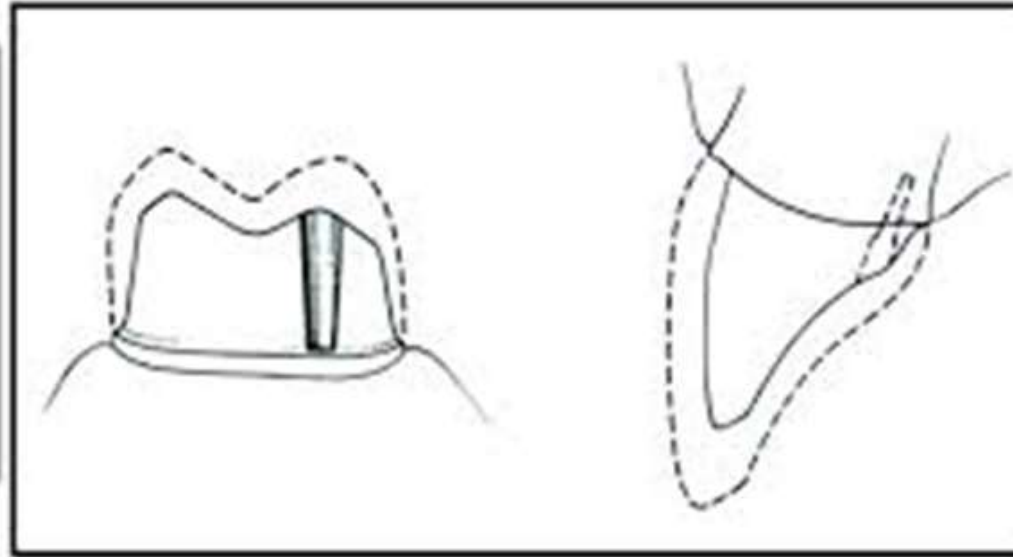
4- Extra retention means : Boxes, grooves, and pinholes may be used to increase retention when needed.



PINHOLES



Extra retention means



RETENTIVE GROOVE



Extra retention means

A



B



C



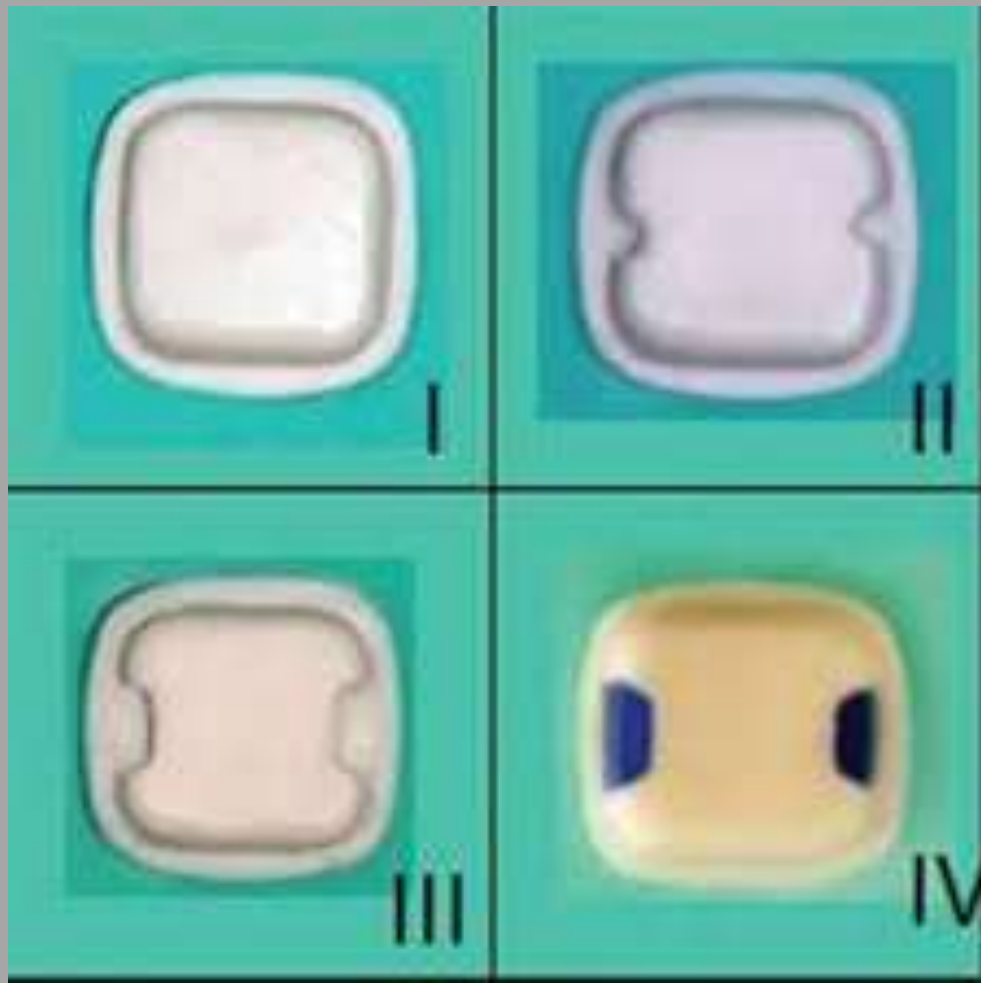
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Retention boxes

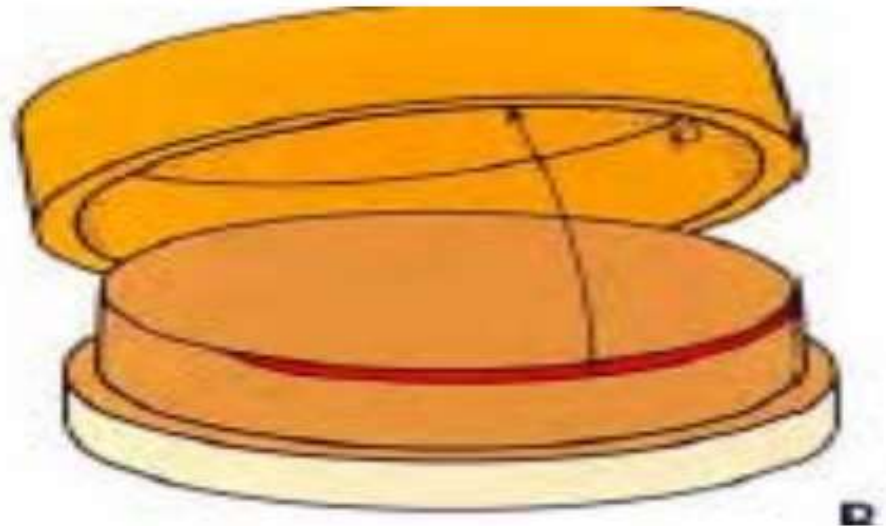
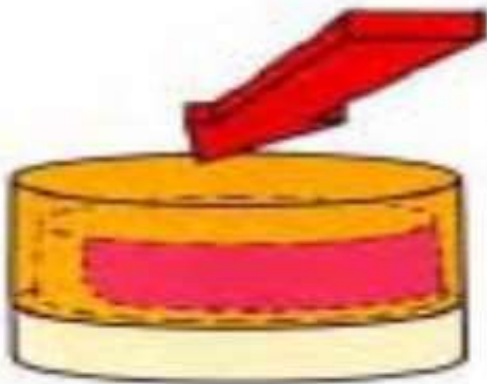
Extra retention means

Retention boxes



4-Diameter of the tooth (tooth width)

Under some circumstances, a crown on a narrow tooth can have greater resistance to tipping than the one on a wider tooth. This occurs because the crown on the narrower tooth has shorter radius for rotation resulting in a lower tangent line and a larger resisting area.



5- height-width ratio:

when there are two crown preparations with similar height (length) the narrower one will be more retentive.

For short preparations it is advised to make the preparation narrower to be more retentive.

This is really what retentive grooves or boxes do when they increase retention.

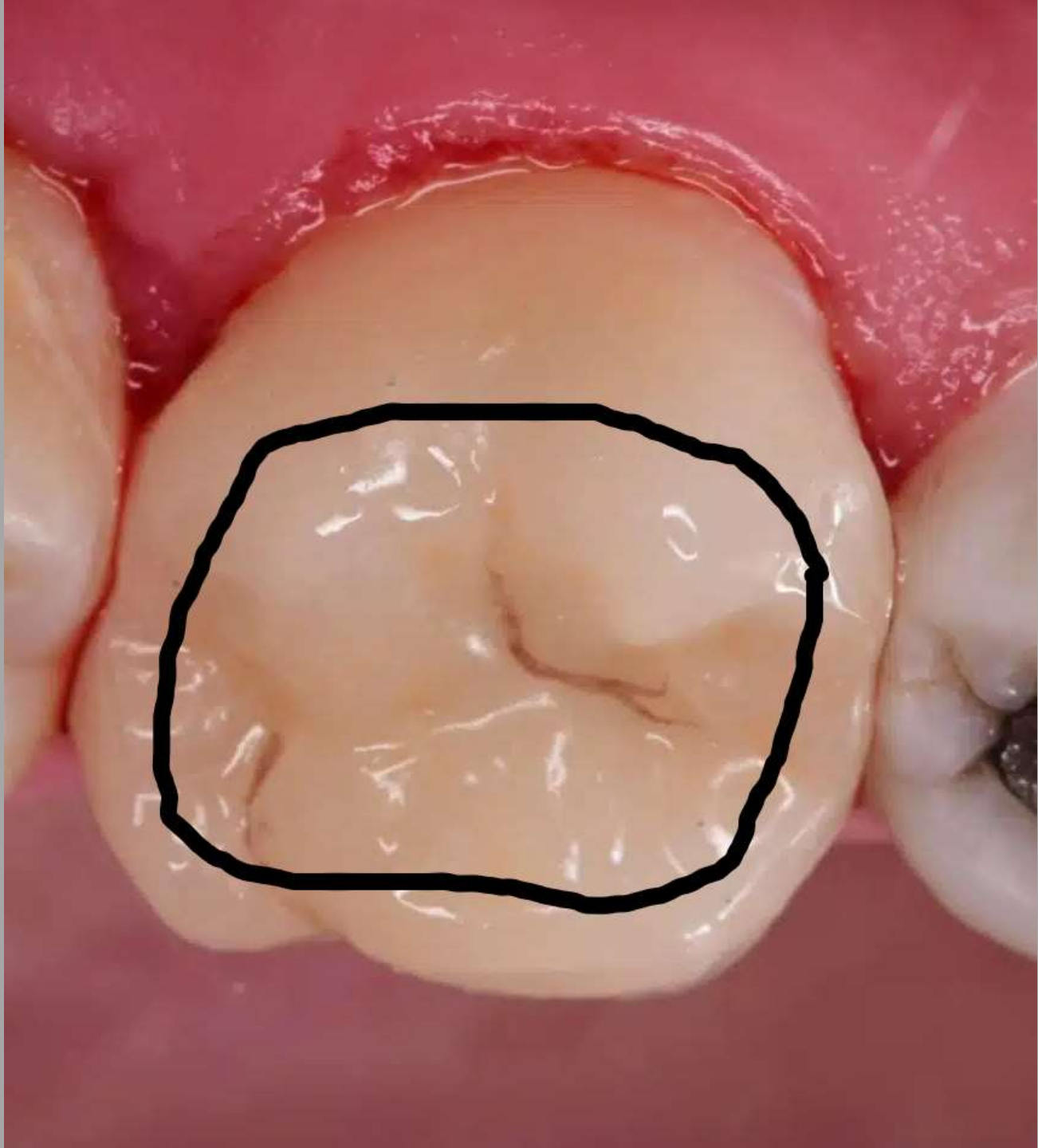
6- post preparations inside the root (post crown): details will be discussed in the next lectures.

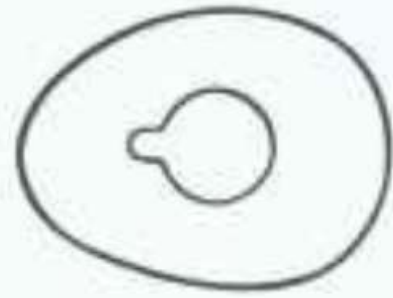
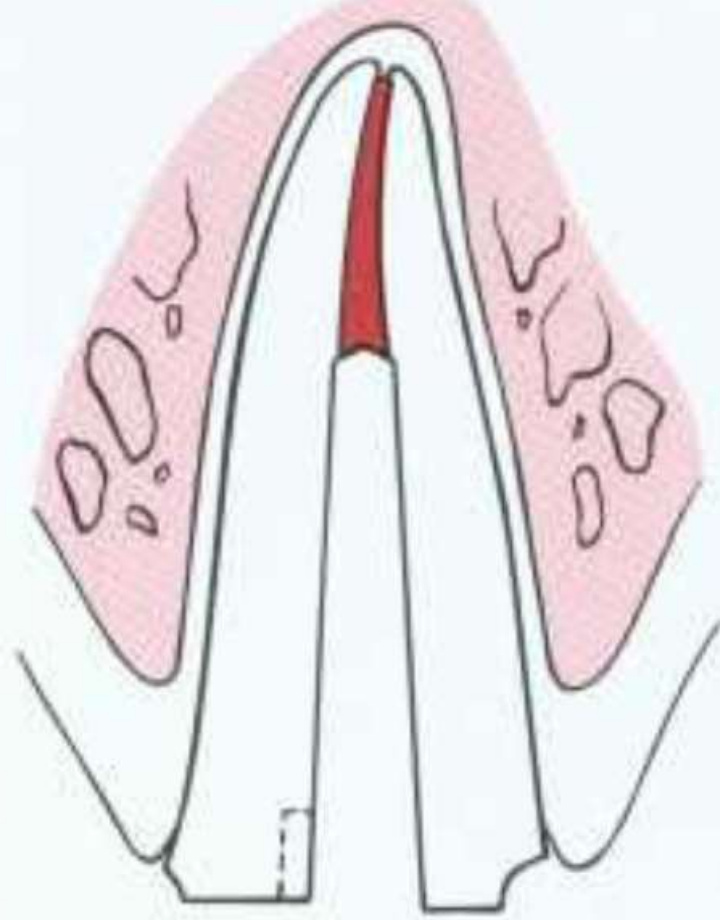
7-Texture of the preparation

Depending on the type of luting cement, the texture of the preparation might affect the retention of cast crown. Smooth surfaces are less retentive than the rough surfaces (mechanical interlocking).

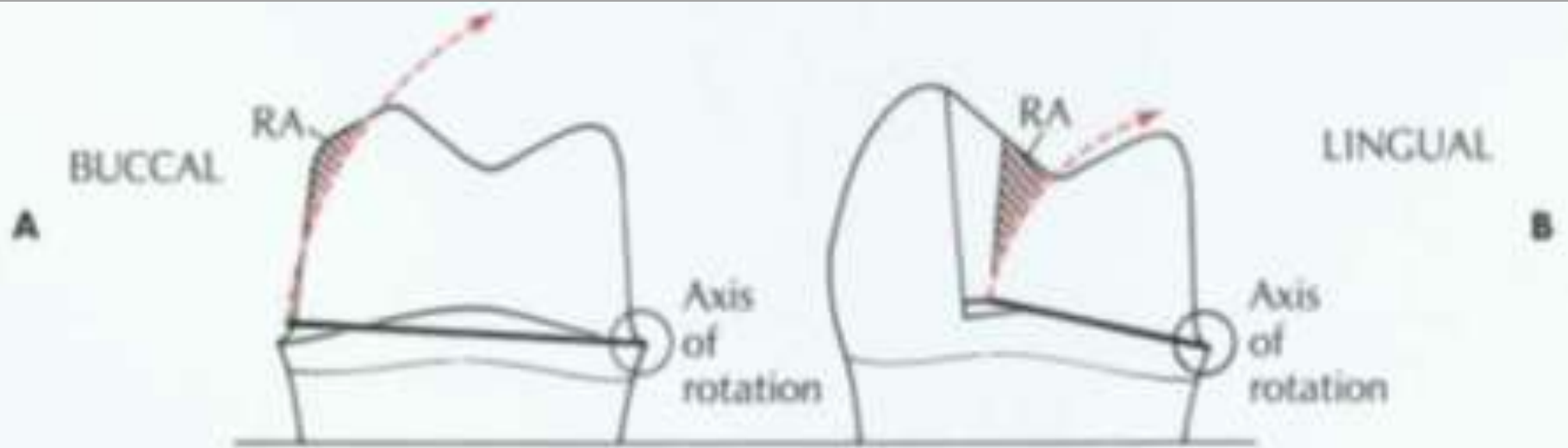
8- Cross-section of the preparation:

During preparation try to follow the same geometrical cross-section of the tooth and avoid round preparations to prevent rotation of restoration.



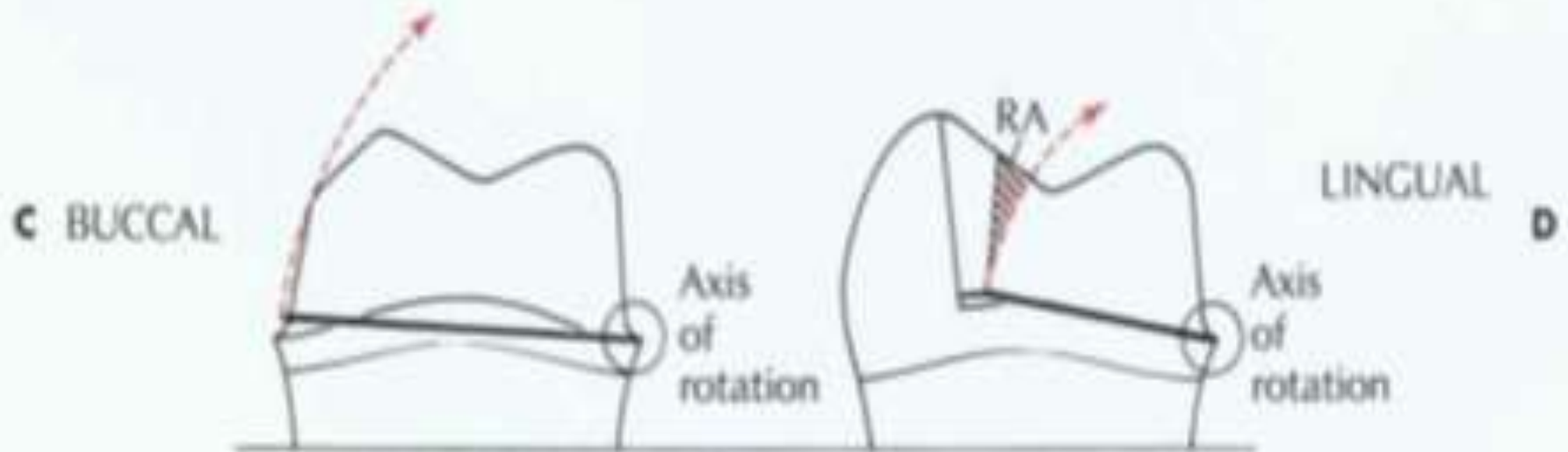


Resistance form in full crown and partial crown.
(**resistance areas RA**) of a complete crown should provide good resistance to rotation of crown around a lingual axis. while in partial crown, resistance must be done by mesial and distal grooves.



In short or excessively tapered complete crown, resistance form is minimal because most of the buccal wall is missing, a mesiodistal groove should be placed to increase resistance form.

Poor resistance form is less probable in a short partial crown because of presence of mesial and distal retentive grooves.



3- Structural Durability:

The preparation must be designed so that it can provide **structure** of the restoration that stay for long **duration**. i.e. the crown restoration must be rigid enough not to flex, perforate (if made of metal) or even fracture (if made of ceramic material).

3- Structural Durability:

For the restoration to be rigid it needs bulk. To provide enough bulk to the crown restoration, sufficient tooth structure must be removed from the prepared tooth to create enough space.

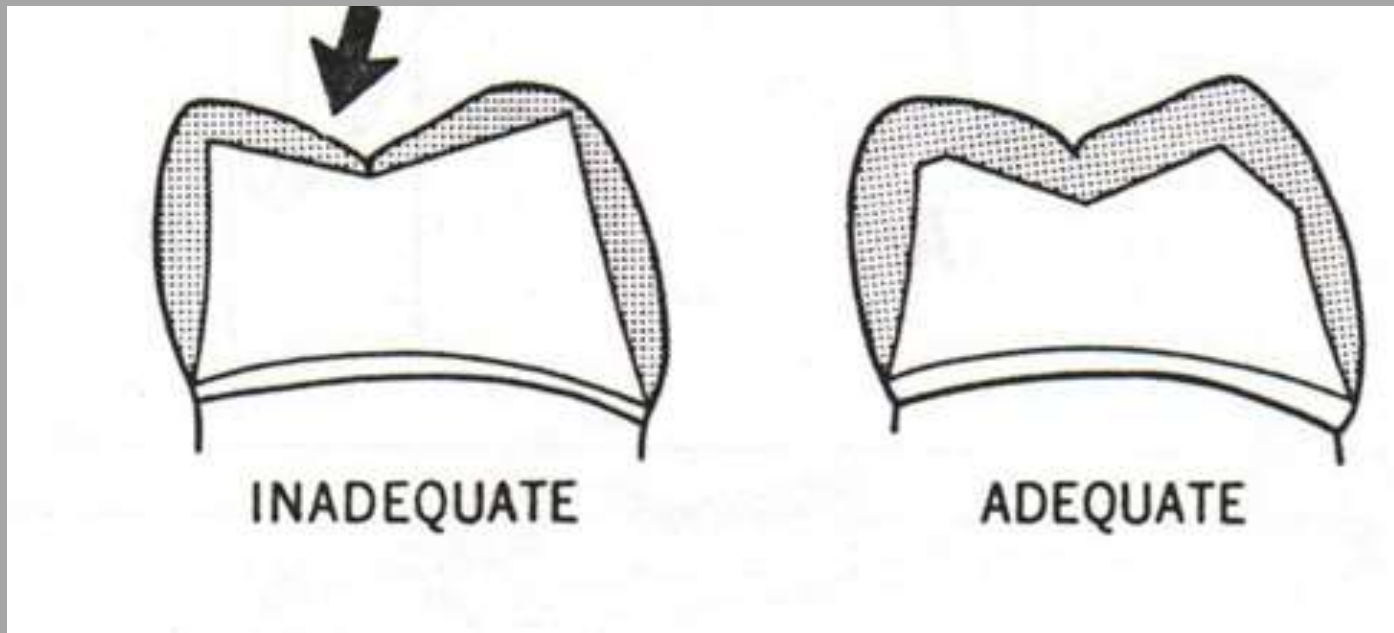
By doing so, the restoration will be allowed to withstand the forces of occlusion, preventing wearing holes in the metal and allowing proper contouring and carving of occlusal anatomy in the restoration.

The preparation must be designed so that it will be possible to have an adequate bulk (thickness) of crown to withstand the forces of occlusion. It also allow proper contouring of the occlusal surface of restoration .

Preparation features related to structural durability :-

1- Occlusal reduction.

2- Axial reduction.



Occlusal clearance:

is the distance (space) between the occlusal surface of the prepared tooth and that of apposing tooth.

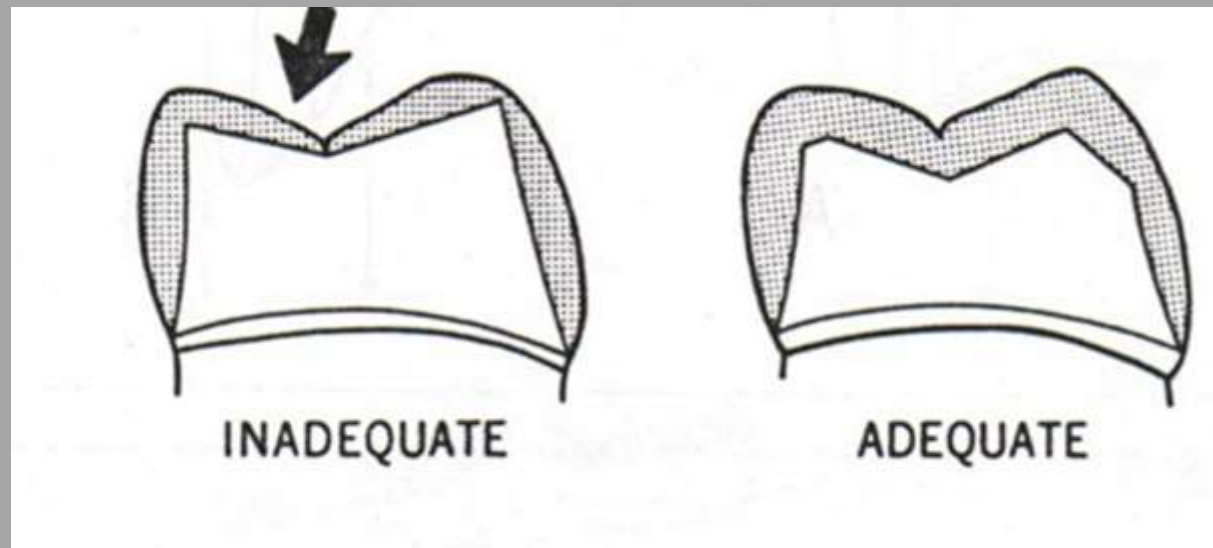
It should be evaluated in centric and eccentric relations.

It should follow the same tooth anatomy this will provide adequate clearance without excessive tooth reduction. A flat occlusal preparation will result in either (1) insufficient clearance or (2) an excessive amount of reduction.



Occlusal clearance:

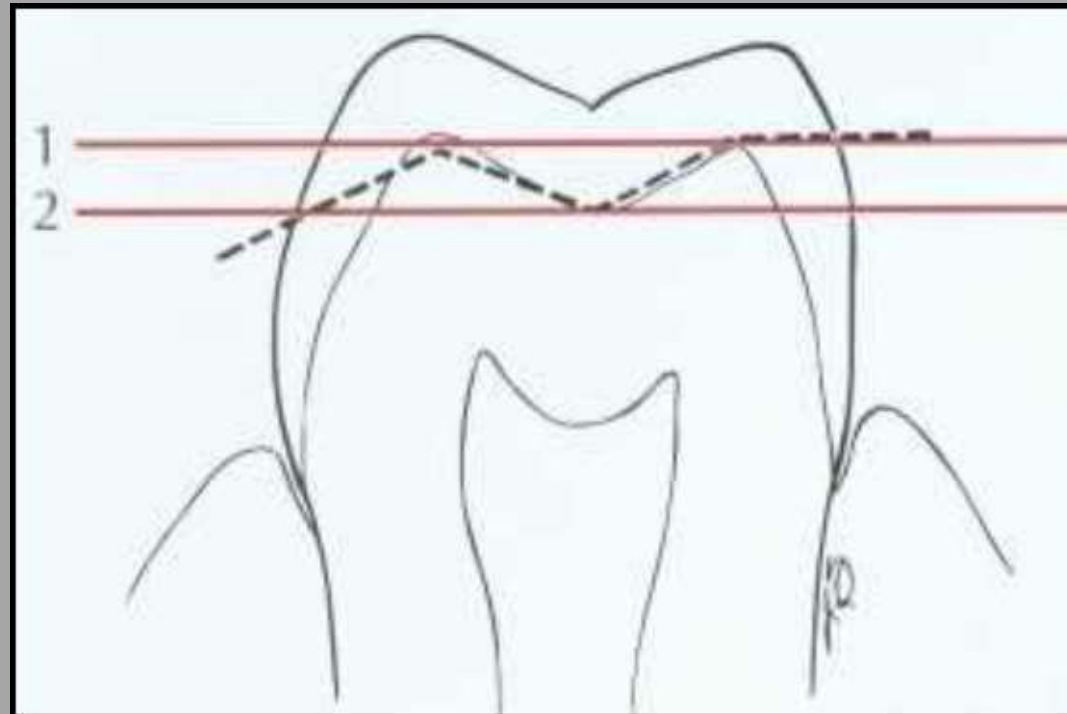
When doing occlusal reduction, we should avoid creating steep planes with sharp angles because it will lead to stress. On the other hand, flat occlusal reduction will lead to too thin metal and this will lead to perforation of the crown restoration in the future.



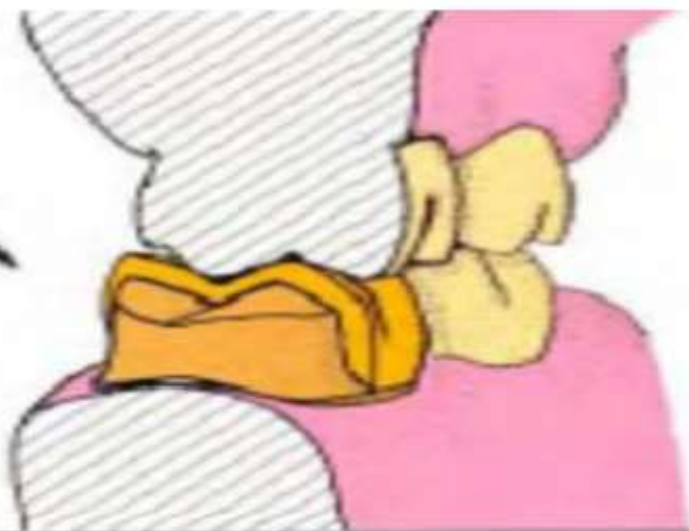
Occlusal clearance:

lowering the entire occlusal surface in an attempt to provide sufficient space might **lead to tooth structure destruction** (non-conservative preparation) which interferes with the conservation of tooth structure.

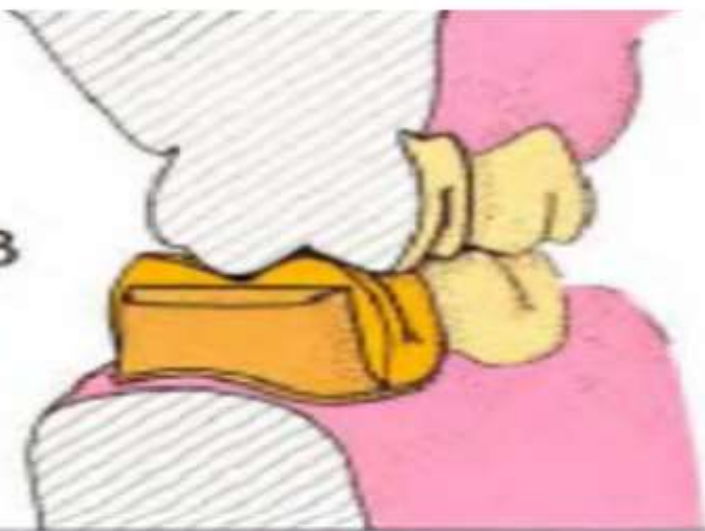
In addition, lowering the entire occlusal surface will **shorten the axial walls** of the prepared tooth which will affect the retention-resistance features of the preparation.



A

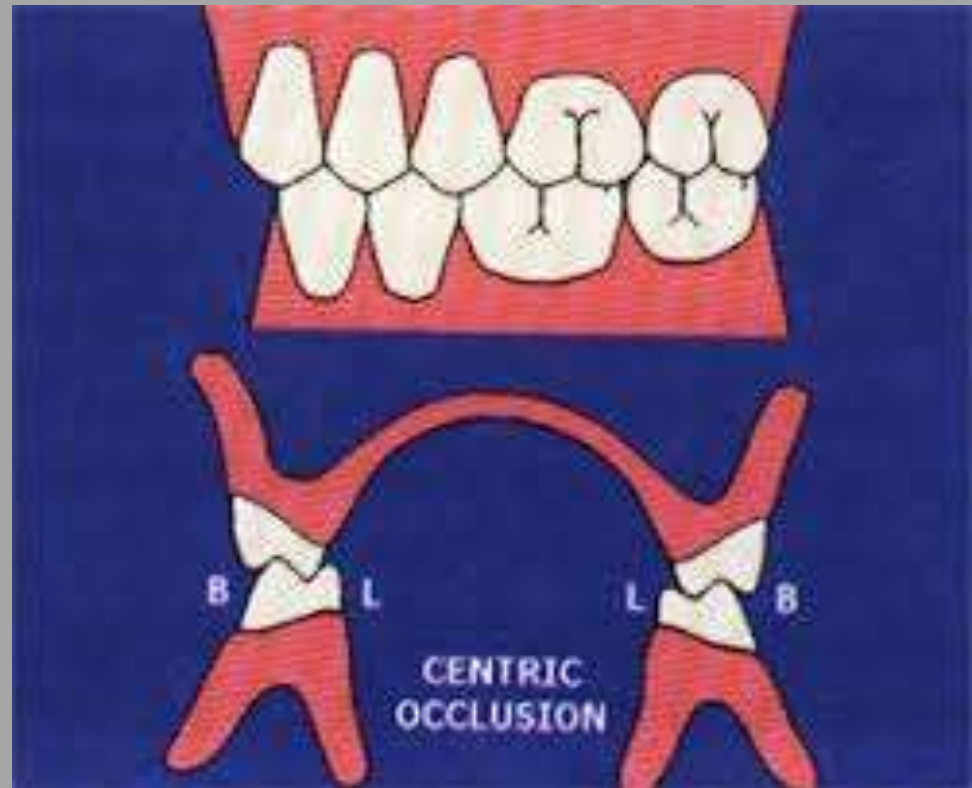
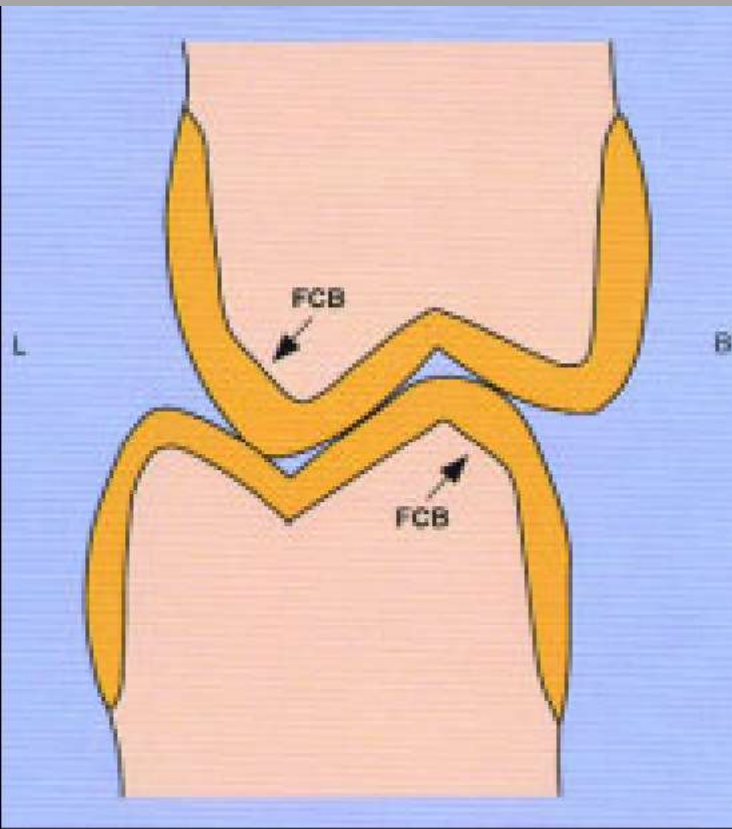


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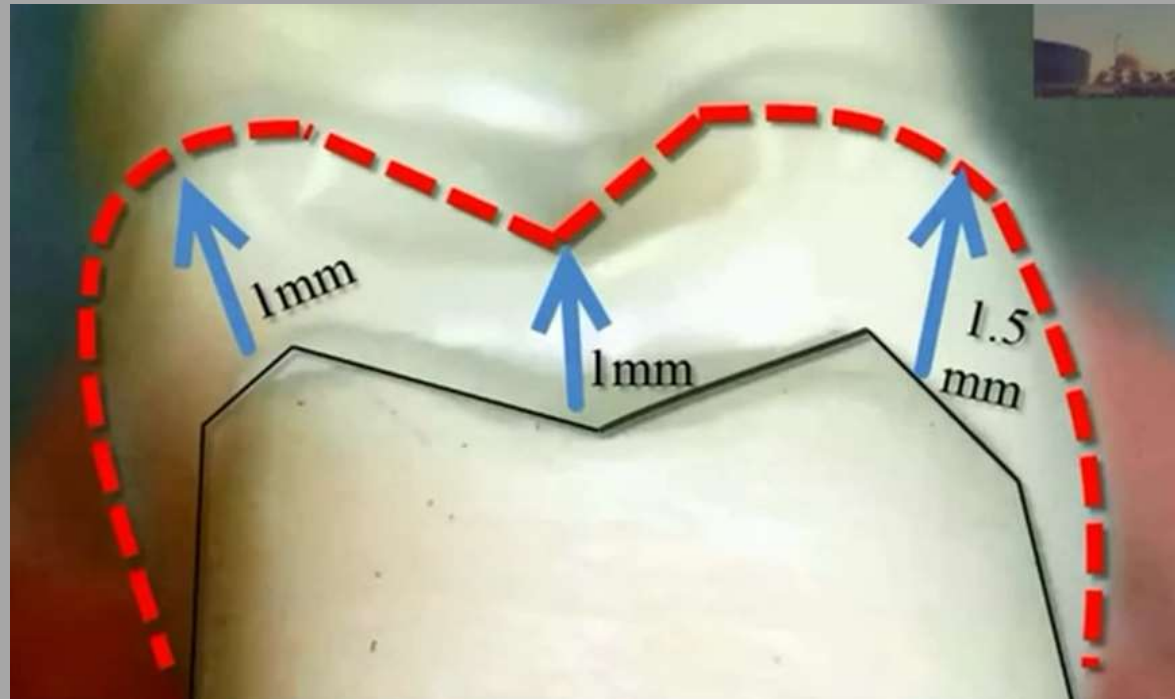
Functional cusps:

The cusps that give centric stops of occlusion (Palatal of upper posterior teeth and buccal of lower posterior teeth). These cusps must have sufficient reduction (1.5 mm) due to heavy forces on these cups.



Functional cusp bevel: (FCB)

is a wide bevel placed on the functional cusps of posterior teeth to provide structural durability. It allows adequate thickness of restoration at this critical area without undue scarifying of tooth structure. If FCB is omitted, the restoration is likely to be too thin in this stress bearing area.



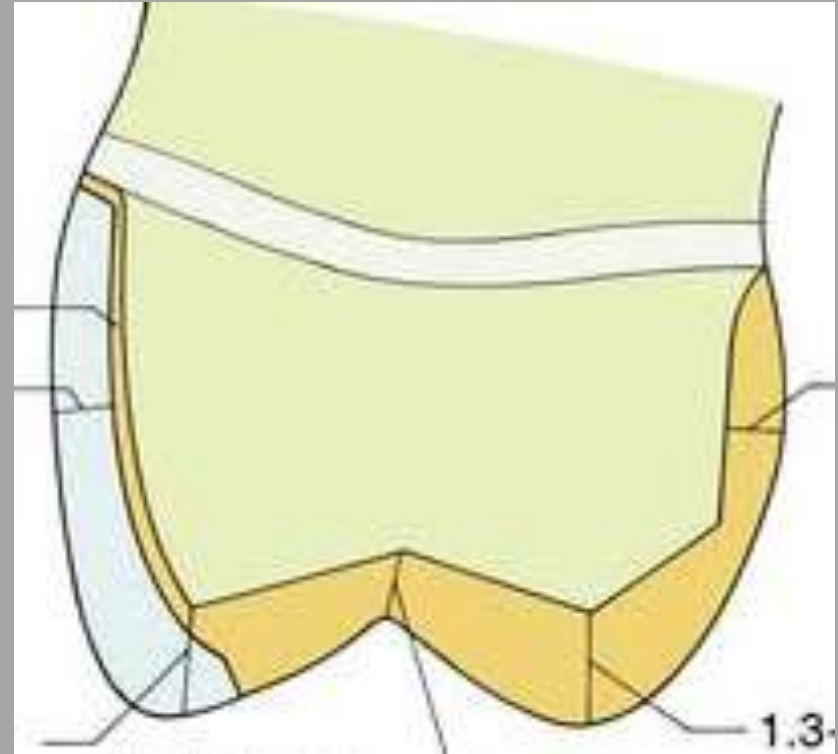
Functional cusp bevel: (FCB)

In the absence of FCB, the laboratory technician overbuilds the crown restoration in attempt to provide structural durability for the restoration; this will lead to premature contact with the opposing tooth.



Axial reduction:

Sufficient axial reduction is important to provide sufficient space so that the restoration can be built with sufficient thickness. This will prevent flexing of the crown restoration when the occlusal forces act on restoration.



4- Marginal Integrity:-

we must do crown preparation so that the restoration margins:

- a-**They must fit as closely as possible against the finishing line of preparation (**clear and smooth finishing line**).
- b-**They must have sufficient strength (**enough reduction**).
- c-** Whenever possible they should be placed in an area where the dentist can finish them properly and the patient clean them properly.

5- Preservation of periodontal tissue:-

1- Whenever possible the margin of the preparation should be supragingival (hygienic position or self-cleansing).

2- The crown should have (enough and proper reduction following the anatomy of tooth structure:

- Proper contact with the adjacent teeth.**
- Enough embrasure form.**
- Proper occlusion with opposing teeth (no premature contacts).**
- Normal anatomical occluso-gingival contour.**



THANK YOU

ANY QUESTION